cloudspaces

CloudSpaces: Open Service Platform for the Next Generation of Personal Clouds

Pedro García López

Universitat Rovira i Virgili

SEVENTH FRAMEWORK PROGRAMME

UNIVERSITAT ROVIRA I VIRGILI

From the Personal Computer to the Personal Cloud

The Personal Computer implied that users had absolute control of their device's hardware and software.

But this recent multi-device approach clearly implies access to Cloud resources that may compromise the privacy and control of user's information.



From PC to Personal Cloud



From the Personal Computer to the Personal Cloud

Forrester: The Personal Cloud: Transforming Personal Computing, Mobile, And Web Markets

Gartner: Top 10 Strategic Technology Trends for 2013

Decisive turning point for european software and infrastructure providers

Google Drive

SkyDrive

Dropbox



CloudSpaces

- Next generation Personal Clouds
  - 3S (**Store, Sync, Share**)
  - **Privacy**
    - Users retaking control of their information
    - User centric approach
  - **Interoperability**
    - Avoid vendor lock-in
    - Enabling third-party applications

UNIVERSITAT ROVIRA I VIRGILI

ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

EURECOM
Sophia Antipolis

CANONICAL

eye os

Tissat

Objectives

- Scalable Data Management
- Privacy-aware data sharing
- Interoperability of Personal Clouds
- Standard service front-end
- Validate and disseminate the platform contributing to Open Source projects

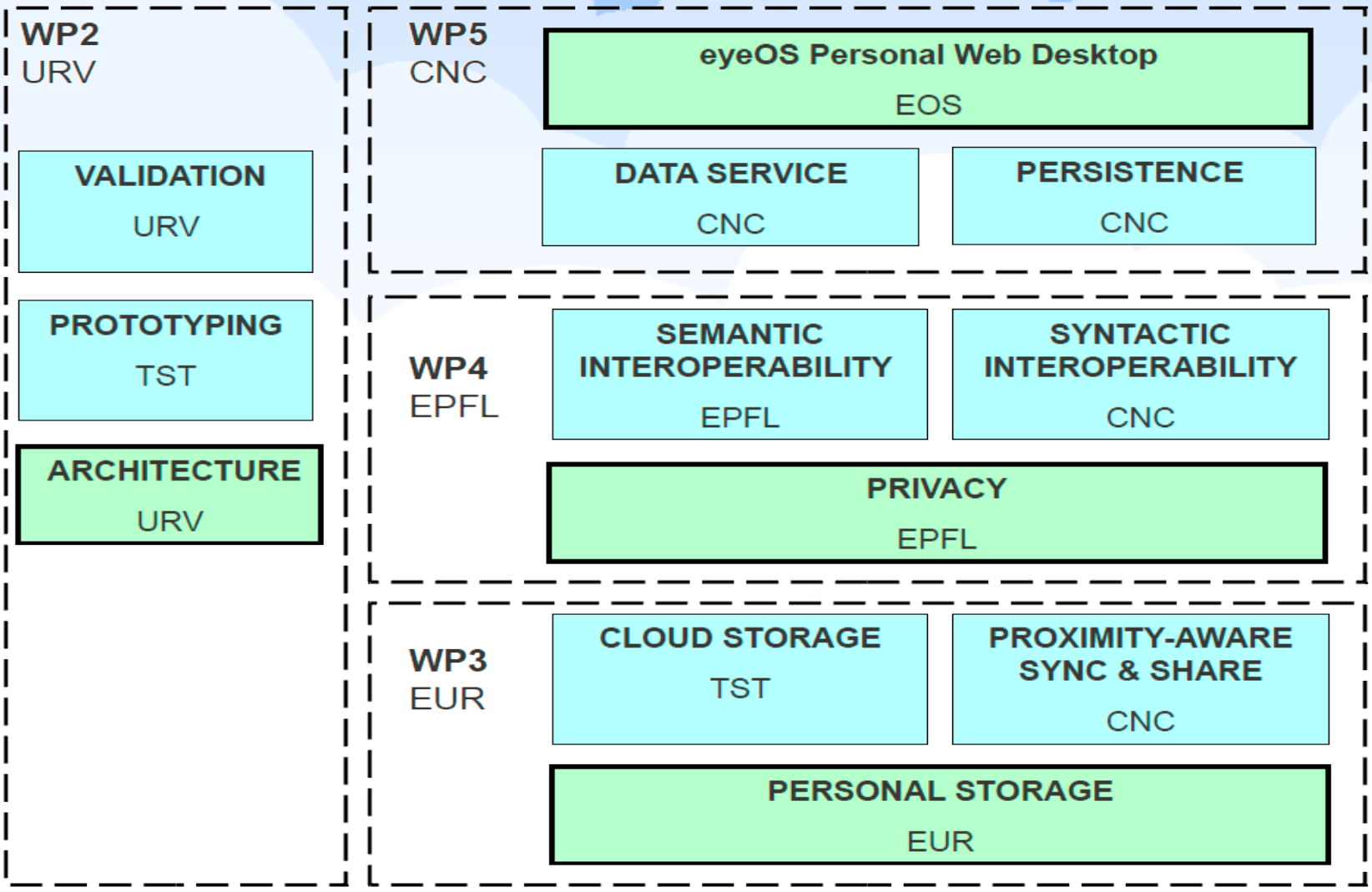




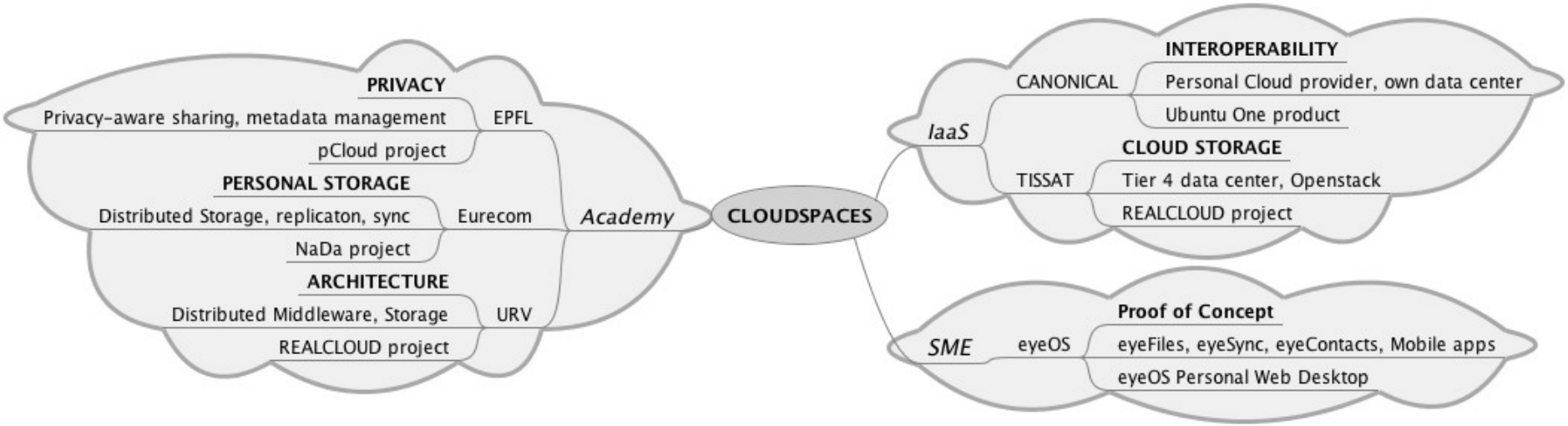




WP Structure



Consortium


CLOUDSPACES
Academy
PRIVACY
Privacy-aware sharing, metadata management
EPFL
pCloud project
PERSONAL STORAGE
Distributed Storage, replicaton, sync
Eurecom
NaDa project
ARCHITECTURE
Distributed Middleware, Storage
URV
REALCLOUD project
IaaS
INTEROPERABILITY
CANONICAL
Personal Cloud provider, own data center
Ubuntu One product
CLOUD STORAGE
TISSAT
Tier 4 data center, Openstack
REALCLOUD project
SME
Proof of Concept
eyeOS
eyeFiles, eyeSync, eyeContacts, Mobile apps
eyeOS Personal Web Desktop

High Impact Results

- Open Service Platform
  - CloudSpaces Storage
  - CloudSpaces Share
  - CloudSpaces Services
- Open Source Projects
  - Ubuntu One
  - OpenStack
  - StackSync
  - eyeOS

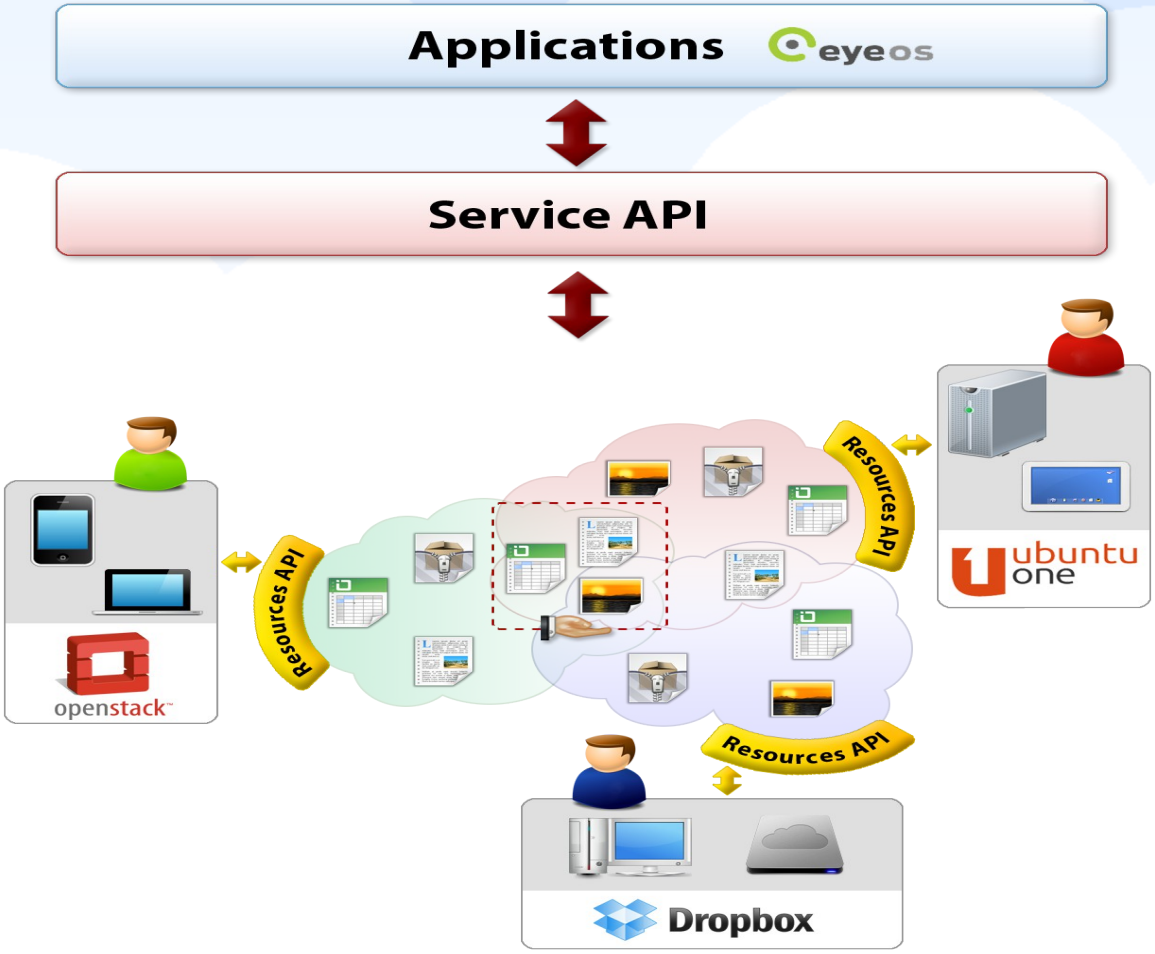




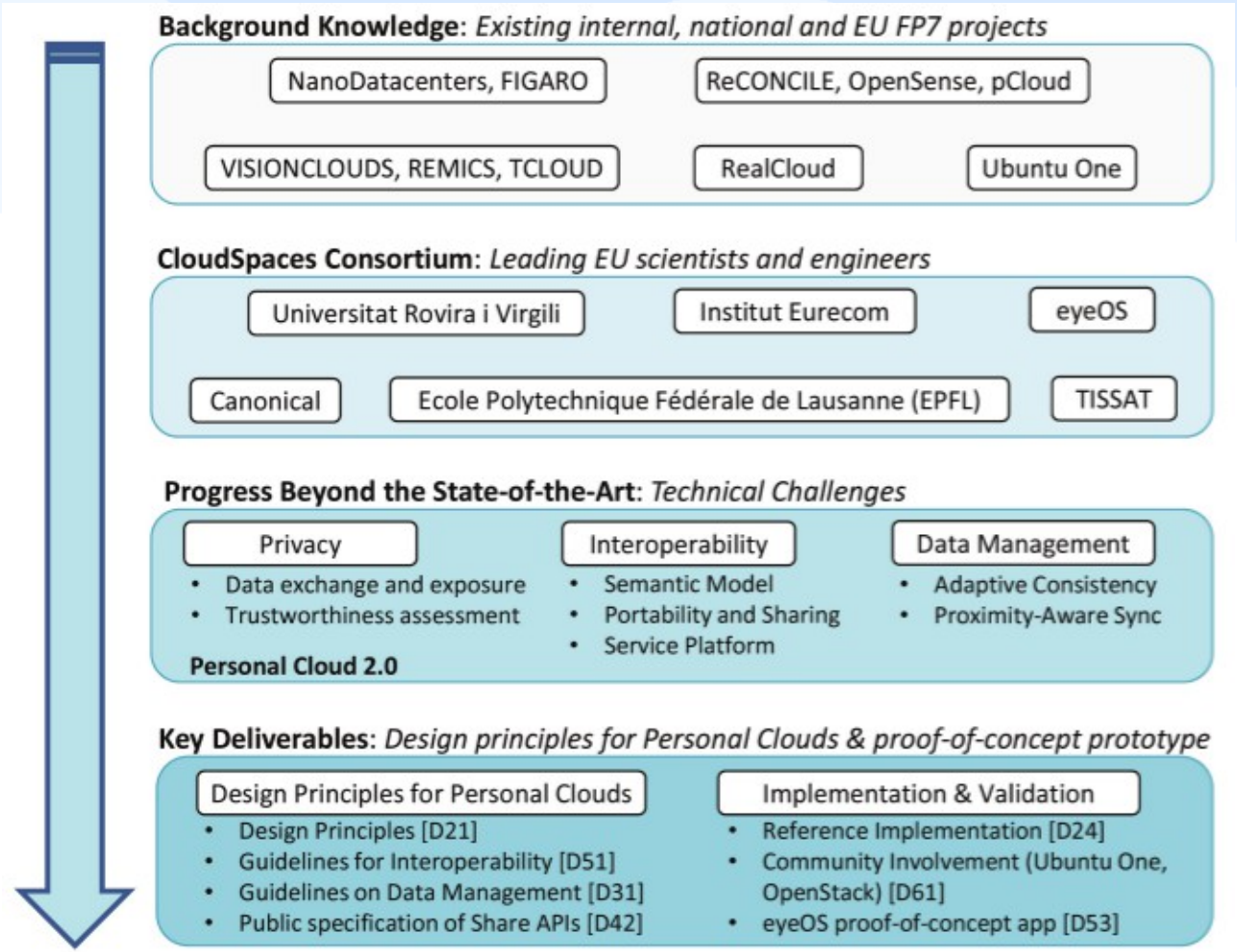




Interoperability



Big picture


Background Knowledge: Existing internal, national and EU FP7 projects
NanoDatacenters, FIGARO
ReCONCILE, OpenSense, pCloud
VISIONCLOUDS, REMICS, TCLOUD
RealCloud
Ubuntu One
CloudSpaces Consortium: Leading EU scientists and engineers
Universitat Rovira i Virgili
Institut Eurecom
eyeOS
Canonical
Ecole Polytechnique Fédérale de Lausanne (EPFL)
TISSAT
Progress Beyond the State-of-the-Art: Technical Challenges
Privacy
Data exchange and exposure
Trustworthiness assessment
Interoperability
Semantic Model
Portability and Sharing
Service Platform
Data Management
Adaptive Consistency
Proximity-Aware Sync
Personal Cloud 2.0
Key Deliverables: Design principles for Personal Clouds & proof-of-concept prototype
Design Principles for Personal Clouds
Design Principles [D21]
Guidelines for Interoperability [D51]
Guidelines on Data Management [D31]
Public specification of Share APIs [D42]
Implementation & Validation
Reference Implementation [D24]
Community Involvement (Ubuntu One, OpenStack) [D61]
eyeOS proof-of-concept app [D53]